

My teaching philosophy is centered on meeting students where they are, building intuition before complexity, and creating a supportive environment for active learning and growth. I believe students learn best when they understand why a concept matters, feel comfortable engaging with difficult material, and gradually take greater ownership of their learning. Whether in classroom instruction or research mentoring, I aim to make challenging ideas approachable without reducing their rigor and to help students develop the confidence and judgment to approach unfamiliar problems independently.

Student-centered learning environment: I recognize that students enter a classroom with different academic backgrounds, prior knowledge, and learning needs. Therefore, I do not expect a single explanation or pace of instruction to work equally well for everyone. I try to understand what students already know and adjust the level of explanation and support accordingly. This approach became more deliberate during my teaching-assistant experience at Iran University of Science and Technology, where I supported courses in Principles of Financial Engineering and Financial Risk Management, conducted problem-solving sessions, and occasionally taught course material directly. When working through quantitative problems, I would revisit foundational concepts when needed and then build progressively toward more advanced steps rather than assuming that every student shared the same starting point. These experiences strengthened my belief that student-centered teaching begins by recognizing where learners are and providing the foundation they need to move forward successfully.

Building intuition before complexity: I believe students understand technical material more deeply when they first develop an intuitive picture of how a method works before confronting its full mathematical or computational complexity. In quantitative and computational courses, students can learn to execute a procedure, apply a formula, or run a model without fully understanding what is happening underneath. I therefore try to make the underlying mechanism visible before moving to greater abstraction or implementation. During the USDA-supported REEU program, for example, I introduced neural networks through an in-person session in which we manually worked through weights, biases, and the basic mechanics of a network before moving to Python and model training. We then used that foundation to discuss concepts such as overfitting and underfitting. The students' reactions were especially rewarding. One student noted, *"I liked a lot the in-person pseudo-class we did where we calculated weights and biases. I felt like you explained overfitting and under fitting well."* Another reflected, *"I had tried to learn how deep learning works before, but for the first time I can say that I understand it."* These moments reinforce my belief that when students first understand the logic beneath a complex method, the mathematics and implementation that follow become far more approachable.

Transparency, accessibility, and trust: I believe that a supportive and open relationship between an instructor and students is essential for effective learning. Students should feel comfortable asking questions, seeking clarification, discussing difficulties, and sharing their progress without feeling intimidated. I therefore make a deliberate effort to be accessible and approachable while maintaining high expectations for their work. This has been especially important in my mentoring, where some of the most productive conversations begin when a student simply stops by my office to ask a question, discuss a challenge, or seek advice. One undergraduate mentee described this experience in a way that strongly reflects the environment I strive to create: *"I mean it with the utmost sincerity when I say that you were beyond helpful during my experience at the REEU program. Not only did you make yourself constantly available and excited to provide any insight or advice on a topic, but you did it in a way that was both efficient and encouraging. While you were incredibly helpful with teaching concepts and answering questions, I think the aspect that had the greatest impact on your ability to help this summer was your willingness to connect to those you were mentoring."* He went on to recall our first meeting, when I stayed with the students after the formal discussion to learn more about them and their interests: *"You showed real interest in who we were not just as scholars, but as people as well, and it is this ability to connect to those whom you are mentoring or teaching that I think will make students excel in your classes and that will make you an excellent teacher/professor."* His words capture what accessibility means to me: not simply being available to answer technical questions, but building the trust that allows students to ask those questions freely and to know that their growth matters beyond a single assignment, class, or research project.

Outcome-driven learning experience: I have always been more engaged in learning when I understand where a concept is leading and why it is worth learning. Reflecting on my own education, I realized that even technically demanding material becomes more meaningful when its purpose is clear. I therefore make a deliberate effort to communicate the learning objective, anticipated outcome, and practical relevance of a topic before becoming immersed in equations, algorithms, or code. Whenever possible, I begin with the problem or decision that motivates the method and then build toward the technical details. This is particularly important in quantitative and computational courses, where students can become proficient in applying a method without fully understanding why or when it should be used. Drawing on my background in industrial and systems engineering, I connect technical concepts to real engineering

and decision-making problems in areas such as manufacturing, healthcare, and intelligent systems. By making the destination visible from the beginning, I want students to understand not only *how* a method works, but also *why it matters and when it becomes useful*.

Active learning and student engagement: I believe students learn most effectively when they become active participants in the learning process rather than passive recipients of information. Therefore, I try to create opportunities for students to apply concepts, make decisions, interpret results, and explain their reasoning. This approach has been especially valuable in my research mentoring. During the USDA-supported REEU program, students moved beyond learning machine-learning concepts in isolation and used them to develop their own research projects, analyze experimental results, troubleshoot unexpected outcomes, and communicate their findings through posters and presentations. I found that these experiences often generated questions and insights that would not emerge through lecture alone, because students had to confront the consequences of their own technical choices. By giving students opportunities to experiment, discuss, and communicate what they learn, I aim to make learning an active process in which understanding develops through participation and ownership.

Lifelong learner: For me, being an effective teacher requires more than expertise in the subject being taught. Teaching itself is a skill that should continually evolve through reflection, feedback, and professional development. An explanation that appears clear from the instructor's perspective may not produce the intended understanding, and an activity that works well for one group of students may need to be reconsidered for another. I therefore value feedback from students, peers, and mentors and intend to use it actively to refine my teaching. I am also interested in continually improving my knowledge of evidence-based teaching practices, assessment, course design, educational technologies, and student mentoring. Just as I expect students to question assumptions, learn from mistakes, and improve their work, I believe an effective educator should model the same willingness to learn.

Teaching interests: I am enthusiastic about teaching undergraduate and graduate courses across three interconnected areas: artificial intelligence and data analytics, optimization and decision sciences, and industrial systems. My primary interests include machine learning and deep learning, computer vision, data analytics, operations research, linear and nonlinear programming, and optimization methods for learning systems. I am also flexible and willing to teach core Industrial and Systems Engineering courses based on departmental needs, including engineering statistics, engineering economy, quality control, operations and supply chain management, and information systems. In addition, I would welcome opportunities to develop specialized courses that bridge modern artificial intelligence and optimization with industrial engineering applications, guiding students from mathematical foundations to computational implementation and ultimately to real-world decision making.

Mentoring: I believe effective mentoring requires knowing when to guide, when to challenge, and when to step back. Having mentored undergraduate, master's, and doctoral students, I have learned to adjust the level of support to each student's experience and stage of development. Early in a project, a student may need examples, structured guidance, and frequent feedback; as confidence and competence grow, I increasingly encourage the student to formulate questions, evaluate alternatives, interpret results, and make independent decisions. My goal is not to make students better at following my instructions, but to make my instructions progressively less necessary. One undergraduate mentee captured this balance in his feedback, noting that my teaching and guidance were clear and that I often gave him "resources to learn more additionally on my own." He also valued my guidance in shaping his project into a comparative evaluation rather than simply completing the technical work. I find it most rewarding when students begin taking ownership of their research such as proposing their own ideas, overcoming technical challenges, communicating their findings, and ultimately seeing themselves as capable independent researchers.

Outreach and Service Activities: I am committed to fostering engagement beyond the classroom through outreach and service activities. For instance, during Mississippi State University's Engineering Discovery Day in Fall 2025, I arranged a session demonstrating how artificial intelligence and computer vision can support industrial quality monitoring through a portable real-time wood-chip assessment system. This activity was part of a K–12 outreach initiative aimed at introducing high school students to engineering disciplines and real-world applications of emerging technologies. Additionally, I have served as an evaluator for science fairs and undergraduate research symposiums, where I assessed student projects and provided constructive feedback. These experiences have strengthened my commitment to encouraging young learners and supporting the development of future engineers and researchers.

Ultimately, I want students to leave my classroom or research group with more than technical knowledge. I want them to be confident in their ability to learn unfamiliar material, ask meaningful questions, apply what they know to real problems, and make independent decisions. If students become progressively less dependent on my guidance and more confident in their own judgment, I consider that a meaningful measure of my success as an educator and mentor.